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Models and Tools for Cyber-Physical Systems
Exercise sheet 11

Exercise 1: Parking a Car

Consider the car model from the course

$$\frac{d}{dt}x(t) = v(t) \qquad\qquad \frac{d}{dt}v(t) = -\frac{k}{m}v(t) + \frac{1}{m}F(t),$$

where x is the position, v the velocity and F the force applied in case of mass m and friction coefficient k. Starting at position $x(0) = -100$, we wish to bring the car to the point 0. To this end, we want to construct a proportional controller K_p for $m = 1000$ (kg) and $k = 50$.

(a) Assuming that we have no feedforward control and measure only the position of the car, provide a continuous-time component for the parking model in block diagram form.

(b) Create a continuous-time component of the proportional controller in block diagram form and indicate how the controller component is connected to the car model component. Finally, combine them together into a single continuous-time component.

(c) Using your numerical solver from last sheet, tune your controller, i.e., find appropriate values for K_p, such that the car is parked at $x = 0$ within $t = 100$ (s). Describe the quality of your controller in terms of steady state error, settling time, overshoot and rise time.

(d) Derive a formula for the Proportional-Derivative (PD) controller that brings the car to the state $(x, v) = (100, 0)$. Here a PD controller is a PID controller whose integral component has coefficient zero, i.e., $K_i = 0$. Moreover, implement the PD car controller using your numerical solver and tune the values of K_p and K_d. Describe the quality of your controller in terms of steady state error, settling time, overshoot and rise time.